

Kubernetes Best Practices — Production Environments

Nexlify Technologies | Cluster: nexlify-prod-eks | ap-south-1 | SLO-bound workloads

1. Production Design Goals

Production clusters balance cost, availability, and operational safety. The compute mix target is approximately 70% Spot and 30% On-Demand capacity (realised via Karpenter mixed NodePools). Every customer-facing Deployment must define a HorizontalPodAutoscaler and a PodDisruptionBudget. Resource requests include a 10% headroom buffer above observed steady-state usage so that short-lived spikes do not cause throttling before HPA reacts. Topology spread constraints and anti-affinity rules keep replicas across AZs. All images are pulled from the private ECR registry; public registries are blocked by admission policy. The platform enforces these standards via Kyverno policies and periodic compliance scans.

2. Spot + On-Demand 70:30 Strategy

Karpenter NodePools for production list both spot and on-demand capacity types. Weight and priority settings bias toward Spot; when Spot inventory is insufficient, on-demand instances of the same family are launched automatically. Critical system components (ingress controllers, service-mesh control plane, monitoring agents) run on a dedicated on-demand NodePool with taints that application pods do not tolerate. Application pods may land on Spot; their PDBs and HPA ensure that capacity loss is absorbed. The 70:30 ratio is monitored via a custom CloudWatch metric that compares Spot versus On-Demand running instance hours; alerts fire if On-Demand exceeds 40% for more than 4 consecutive hours.

3. Horizontal Pod Autoscaler Requirements

Every Deployment or StatefulSet that serves external traffic must have an HPA. Minimum replicas is at least 2; maximum is set according to capacity planning (default 10 unless justified). Metrics may be CPU, memory, or custom (e.g., requests-per-second via Prometheus adapter). Scale-up stabilisation is 60 seconds; scale-down is 300 seconds to avoid flapping. HPA behaviour is validated in staging before promotion. For workloads with very spiky traffic, KEDA may be used in addition to HPA, but only after Architecture Review Board approval.

4. PodDisruptionBudget Standards

A PDB is mandatory for every production Deployment with more than one replica. The preferred expression is maxUnavailable: 1 for small replica counts, or minAvailable: 50% for larger sets. PDBs protect against both voluntary disruptions (Karpenter consolidation, node upgrades) and involuntary ones where possible. During a planned cluster upgrade the Platform team temporarily raises disruption budgets after confirming HPA can compensate. Applications that cannot tolerate any disruption (rare) must run on the on-demand system NodePool and document the exception.

5. Resource Requests with 10% Buffer

Resource requests are derived from production metrics (p95 CPU and memory over a 7-day window) plus a 10% buffer. Limits are set higher (typically 1.5–2× requests) to allow bursting. The 10% buffer reduces the chance that a pod is throttled before HPA adds replicas. Vertical Pod Autoscaler is not used in production at this time; requests are adjusted manually during capacity reviews. QoS class should be Burstable for most application pods; Guaranteed is reserved for latency-critical components.

6. Additional Production Controls

- Readiness and liveness probes with production-grade timings (initialDelaySeconds $\geq$ 10, periodSeconds 10).
- PreStop hooks for graceful connection draining (at least 15 seconds).
- NetworkPolicy default-deny with explicit allow rules for required peers.
- Pod Security Standards restricted profile enforced via admission.
- Image digests pinned; tags alone are insufficient.
- Topology spread constraints with maxSkew 1 across zones for replica sets of size $\geq$ 3.

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